
Aldoc: A Framework for Generating and Identifying AI-Edited Documents

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Abstract

1 The rapid proliferation of AI-generated content has created an urgent need for
2 robust detection methods. Despite its ubiquity, document-level detection remains a
3 significant and largely unaddressed research frontier. This gap exists not only due
4 to the absence of specialized detection models for this modality but also because the
5 necessary data to develop them was previously non-existent. In this work, we tackle
6 these two equally critical challenges through two fundamental contributions. First,
7 we introduce **AldocFactory**, a pioneering framework that enables the automated
8 generation of complex, synthetic document datasets, establishing a new foundation
9 for research in this area. Second, we present **AldocDetector**, a Vision-Language
10 Model (VLM) that analyzes documents as holistic entities to accurately detect and
11 segment AI-generated regions. Experimental results demonstrate the effectiveness
12 of our methodology, achieving a Dice score $> 85\%$. By releasing our code and
13 datasets, we provide the first end-to-end open-source ecosystem for document
14 integrity, opening a new path for the community in the age of generative AI ^{1 2}.

15 1 Introduction

16 AI-generated content (AIGC) has become an escalating crisis across domains. Synthetic content
17 and automated bots now dominate a significant portion of global web traffic [30]. This has led to
18 the “AI slop” phenomenon—low-quality, AI-produced content flooding online platforms to such an
19 extent that it was named Merriam-Webster’s 2025 Word of the Year [18, 40]. Beyond mere clutter,
20 the “Dead Internet Theory” warns of a near future where the erosion of authenticity renders the web
21 functionally useless [42]. As distinguishing real from AI-generated material becomes increasingly
22 difficult, the scientific community faces a critical challenge in safeguarding digital trust [51].

23 In response, the field of AIGC detection has experienced exponential growth. Historically, forensic
24 research has closely mirrored the rapid trajectory of generative AI development. Early efforts focused
25 on mitigating the risks of “deepfake” facial manipulation—one of the first generative milestones
26 to achieve alarming levels of realism and necessitate the immediate development of verification
27 technologies. As synthesis capabilities matured, the scope of detection pivoted to encompass general
28 image generation and high-quality synthetic text produced by Large Language Models (LLMs).
29 However, as generative architectures continue to evolve, detection strategies must not only refine
30 existing methodologies but also expand their reach to encompass increasingly complex modalities.
31 While recent advances have begun to address formats such as audio, video, source code, or tabular
32 data, a ubiquitous yet critical data type has been systematically overlooked: the document.

33 In recent years, the rapid evolution of AI, coupled with the pervasive role of documents in modern
34 life, has catalyzed the emergence of Document AI [12, 28]. This field has produced models capable

¹<https://anonymous.4open.science/r/AldocFactory-C301>

²<https://anonymous.4open.science/r/AldocDetector-A637>



Figure 1: **Overview of our dual-pillar solution.** First, AIdocFactory creates tailored training datasets by generating and editing documents to overcome the current absence of data. These complex datasets power AIdocDetector, our novel Vision-Language Model. By projecting the documents into a dedicated embedding space, the detector successfully exploits structural and semantic signals to accurately localize AI-generated content (AIGC).

of advanced analysis, understanding, and verification, as well as the manipulation of the documents themselves. The malicious deployment of these capabilities poses severe societal and economic threats; for instance, AI-generated scientific papers can pollute the academic record and misdirect future research, while fabricated legal contracts or financial records can lead to catastrophic fraud. Parallel to the advancements in other data modalities, it is essential to develop AIGC detection frameworks in documents that evolve in tandem with these emerging manipulative capabilities.

To address these escalating challenges, we propose a dual-pillar solution that tackles the two fundamental barriers in this field: the absence of tailored datasets and the lack of robust document detection models. First, we present AIdocFactory, an open-source framework for AI-powered document generation and editing. Specifically, AIdocFactory (1) extracts raw document content via Docling [13] and structures it, (2) applies targeted modifications via specialized editing modules, (3) exports the synthesized documents, and (4) creates the final database. This tool is essential for creating the complex, diverse, and large-scale databases required to train next-generation detectors—a resource that was previously non-existent. Second, we introduce AIdocDetector, a novel Vision-Language Model (VLM) specifically designed to detect and localize AIGC within documents. Leveraging a SmolVLM-256 architecture [43], our approach projects the entire document into a continuous embedding space, capturing its holistic representation for high-precision analysis. Together, these components provide a complete end-to-end ecosystem for ensuring document integrity. An overview of the proposed framework is shown in Figure 1.

The main contributions of this work are summarized as follows:

- **A New Forensic Paradigm:** We formally expand the scope of AIGC detection to a critical yet systematically overlooked modality: the document. This approach follows the same trajectory established for other modalities, such as audio, video, and code, and anticipates the necessary expansion to further data types in the future as AI continues to evolve.
- **AIdocFactory & Benchmark:** We introduce a pioneering open-source engine for the automated generation and sophisticated editing of multimodal documents, enabling the community to generate tailored synthetic data at scale. Using this framework, we release the first comprehensive benchmark for document-level AIGC detection, comprising over 35,000 high-quality pages with precise region-level ground truth.
- **AIdocDetector Architecture & Weights:** We propose a novel Vision-Language Model (VLM) engineered for the document domain. By directly encoding the full document into a continuous embedding space, the model naturally exploits its holistic features to provide both robust classification and precise pixel-level segmentation of AI-generated regions.
- **Rigorous Evaluation & Open Ecosystem:** Extensive evaluations confirm the strong detection and localization capabilities of our approach (83% accuracy, 85% Dice score). To foster transparency and future research, we publicly release the complete end-to-end ecosystem: the generation engine, the dataset, and the detection model.

2 Related Work

The detection of AI-generated content (AIGC) has become increasingly critical due to its widespread societal impact. Existing literature can be categorized by the data modalities they target:

75 **Image Generation Detection.** As generation models become increasingly photorealistic, detec-
76 tion techniques must identify increasingly subtle forensic artifacts [63]. Detection methods exploit
77 generator-specific characteristics through frequency and spatial domains analysis [16, 11, 52], repre-
78 sentational robustness to input noise [23], as well as Vision Transformer-based features [24].

79 **Text Generation Detection.** The rise of LLMs has driven extensive research into identifying synthetic
80 text [59], primarily through statistics-based or neural-based methods. Statistics-based approaches,
81 such as Rank [50] or perplexity-based tools like HowkGPT [56] and LLMDet [61], leverage the
82 predictability and statistical properties of language. Conversely, neural-based methods use trained
83 classifiers; while early models relied on handcrafted indicators like syllable count or syntactic
84 complexity [49], contemporary research focuses on fine-tuned transformers—including BERT [14],
85 RoBERTa [38], and XLNet [66]—which consistently outperform statistical detectors [26, 45, 55].

86 **Emerging Modalities.** The continuous improvement of generative models has necessitated the
87 expansion of detection research into increasingly diverse data types. Recent literature has introduced
88 specialized frameworks for synthetic audio [3, 36], video forensics [32], and source code attribution
89 [31]. Furthermore, emerging work has explored synthetic tabular data identification [34] and graph-
90 based detection of botnets and synthetic social networks via non-organic interaction patterns [22].

91 In tandem with these modalities, new frontiers must emerge to mirror the rapid development of
92 generative capabilities; for instance, AIGC detection for web-scale infrastructures will become
93 essential to identify increasingly prevalent AI-generated websites. Our work fills the gap in documents
94 by proposing the first document-level AIGC detection framework that operates directly on the
95 document domain for both classification and segmentation.

96 3 Background and Motivation

97 Documents constitute the bedrock of the global information ecosystem. With an estimated trillions
98 in active circulation, they represent the second most common file type on the web after HTML,
99 accounting for a massive portion of indexed content [2, 10].

100 **Document AI as a Holistic Entity.** Beyond their ubiquity, a document is a “structured unit of
101 information that combines textual, layout, and visual elements to convey a specific communicative
102 intent” [7, 8]. Traditionally, Document AI [12, 28] relied on fragmented pipelines: OCR extracts
103 raw text, followed by an ensemble of independent tools for text processing, layout analysis, table
104 extraction, image analysis, and numerous other specialized modules. However, the advancements in
105 Document AI have driven a paradigm shift, treating documents as complex, holistic entities rather
106 than collections of isolated parts [33, 43, 57]. This integrated approach not only preserves critical
107 spatial and semantic relationships—such as image-text coherence—but also consolidates all tasks into
108 a single model to achieve a multi-level understanding of the document. Building on this foundation,
109 we establish a previously unattainable paradigm: AIGC detection on documents. Operating in this
110 native space underpins robust detection and enables the segmentation approach presented in this
111 work, allowing for the precise localization of generated components throughout the entire document.

112 **Detecting Intrinsic Generative Patterns.** It is crucial to distinguish our task from ID forgery
113 detection. While IDs are a subset of documents, their unique characteristics dictate a fundamentally
114 different forensic approach. Designed for the physical realm, IDs incorporate sophisticated security
115 mechanisms—such as watermarks and Optically Variable Devices (OVDs)—specifically engineered
116 to deter forgery. Digitally, these IDs exist primarily through photographs or scans. In this context,
117 forgery detection targets visual artifacts introduced during the manipulation of these digitized captures
118 of physically secured documents. Traditional techniques often rely on high-frequency analysis [15],
119 while advanced frameworks, such as Trufor [21], exploit Photo Response Non-Uniformity (PRNU).

120 However, the task addressed in this work is fundamentally different. We focus on documents that
121 are naturally digital (born-digital). In the born-digital domain, the visual artifacts exploited by ID
122 forensics simply do not exist. Instead, detection must rely on the document content itself, requiring
123 the identification of intrinsic generative patterns through a holistic analysis of the document as a
124 unified information structure. To this end, our approach projects the entire document into a continuous
125 embedding space, enabling the model to capture its holistic representation.

4 AIdocFactory

The primary obstacle in developing this technology is the absolute lack of existing data. Consequently, the first pillar of our work is AIdocFactory, a specialized tool bridging this gap. The system operates through a coordinated processing flow composed of four primary modules, each handling a distinct stage in the document generation pipeline to ensure modularity, scalability, and robustness (Fig. 1).

Docling Parsing. The first module leverages Docling [29] to extract all document components and their attributes—such as titles, paragraphs, images, tables, footnotes, bibliographic entries, and others—compiling them into a structured hierarchical variable (see Appendix A.4). This variable is then transformed into a streamlined format optimized for downstream processing, which we refer to as the *Document Variable*. This representation accurately captures all document components and retains the information necessary to faithfully reconstruct the document later in its original form.

AI Edition. This module modifies specific components within the Document Variable, giving users full control over the process via customizable parameters (see Appendix A.5). Each modification is routed to a dedicated submodule based on the component type. Users can specify any generative model for these modifications, either directly via the Hugging Face identifier [27] or through a custom local model. This flexibility allows for the seamless modification of different components, enabling users to select the most appropriate generative model for each specific component. Detailed operational logic is available in the documentation ¹.

Crucially, this module exclusively modifies the semantic content within the Document Variable (Appendix A.4.8), leaving all structural and layout metadata untouched. By strictly isolating modifications to the content level, we ensure that the only difference between an original and a modified document lies in its content rather than visual artifacts, which are not the scope of this work.

Exporter. The third module takes the AI-modified Document Variable and renders it into a new document. While the framework supports a *basic* mode for predefined HTML-based outputs, the creation of our benchmark exclusively utilizes the *strict* export mode. Implemented via the *fitz* Canvas library [4], this mode faithfully reproduces the structural fidelity of the original document. Crucially, both original and AI-generated components are rendered through the exact same export pipeline. Since only the semantic content is altered and all elements are exported indistinguishably, we guarantee that the sole difference between an authentic and a modified component is its intrinsic content—as corroborated by the experimental results in Sec. 6.5 and Appendices A.2 and A.3.

Database Management. Running in parallel with the export stage, this module handles all SQL database operations. It tracks and stores which components were altered, recording their precise bounding boxes—which serve as essential ground-truth labels for AIdocDetector training—alongside additional metadata and generated content. For instance, it stores raw text, extracted images, or tables, enabling their use in other tasks (Sec. 4.1). This detailed record ensures full traceability and facilitates downstream reuse. An illustrative example of the stored data schema is provided in Appendix A.1.

Overall, AIdocFactory constitutes a versatile and extensible framework that provides high-quality, large-scale data to train and evaluate AI document detectors. The specific configuration used to generate the benchmark dataset for this work is detailed in Sec. 6.1, and a qualitative and quantitative analysis demonstrating the output quality is provided in Appendices A.2 and A.3.

4.1 Use Cases and Extensibility

AIdocFactory’s modular architecture is explicitly designed for maximum extensibility. Below, we present several practical use cases that our framework *already* supports natively by simply selecting and combining its constituent modules. While these functionalities are highly valuable and could constitute standalone tools, we provide them here fully open-sourced:

- **Format Conversion:** Integrating the *Docling* and *Exporter* modules enables multi-format document conversion (e.g., transforming a photographed physical document into a structured PDF). This functionality is readily extensible to support any desired output format.
- **Data Ingestion:** *Docling* and *Database* modules enable the automated analysis of document collections, constructing structured SQL databases with metadata, raw text and images.

- **Dataset Curation:** By combining the *Docling*, *AI Edition*, and *Database* modules, users can extract document components and process them with generative models to produce specialized AI-generated image or text datasets.
- **Synthetic Generation:** The full pipeline can be deployed for controlled document generation tailored to various downstream tasks (e.g., training Document Question Answering (DocQA) models or Spatial Entity Localization).

Furthermore, AIdocFactory is designed for high usability, utilizing a dedicated configuration file that allows users—regardless of programming expertise—to operate and fine-tune the system without requiring modifications to the source code ¹.

5 AIdocDetector

AIdocDetector is a VLM that analyzes the document as a holistic entity to detect and localize AI-generated content. As illustrated in Figure 2, it processes document pages to produce: (1) a binary classification label, and (2) a segmentation mask identifying synthetic regions.

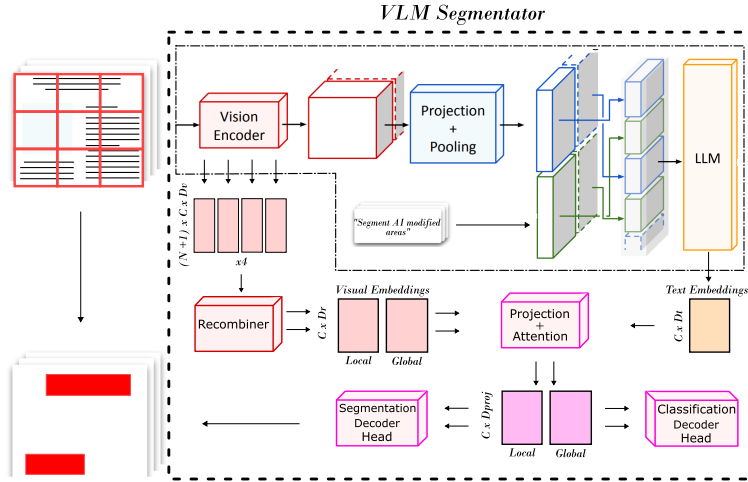


Figure 2: Overview of the AIdocDetector architecture. The model processes document pages holistically, performing dual tasks: global binary classification of AI-generated content, and precise pixel-level segmentation of synthetic regions.

The proposed architecture is the culmination of rigorous experimentation aimed at addressing the unique challenges of document-level AIGC detection. It is important to contextualize that our primary objective is to establish a robust, foundational architecture that successfully demonstrates the viability of this novel detection task. While exhaustive hyperparameter optimization or fine-tuning architectural details could yield marginal performance gains, our focus remains on introducing a principled, end-to-end framework rather than chasing improvements in evaluation metrics. The resulting architecture effectively proves that AIGC detection is achievable at the document level.

Rather than undertaking the expensive pre-training required to teach a model holistic document understanding, we initialize our architecture with the weights of SmolDocling-256 [43]. This directly transfers a strong inductive bias for holistic document analysis. The core of this projection relies on the Vision Encoder, which maps the entire document into a feature space. To further align these representations with our specific task, we apply Low-Rank Adaptation (LoRA) [25] to the encoder, enabling an efficient and highly targeted refinement of these visual features.

To effectively process this representation, we employ a multi-scale feature extraction. While early layers capture global structural layout and hierarchical information, deeper layers encode the fine-grained semantic details. Furthermore, extracting features across four distinct depths allows the segmentation head to precisely localize characteristics within the original image dimensions [39, 62].

Although the Vision Encoder theoretically encapsulates all necessary information, relying exclusively on its dense representations proved insufficient, as analyzing them in isolation demands an overly

complex decoding head. To alleviate this decoding bottleneck, our architecture purposefully adapts the LLM decoder. By integrating and repurposing this decoder into our pipeline, we leverage its decoding capabilities to facilitate the extraction of meaningful features, effectively distributing the representational load. To empirically validate whether merging these two representations is strictly necessary for the task, we present the experiment in Figure 3, which illustrates the loss and Dice score curves when relying exclusively on Vision Encoder features (left) versus solely on LLM decoder features (right). Compared to our combined approach (Figure 7), we observe that not only are the validation metrics significantly worse, but the training loss curves also demonstrate an inability to converge on this dataset. This indicates an inherent lack of fundamental information required to execute the task, confirming that their combination is essential for successful detection.

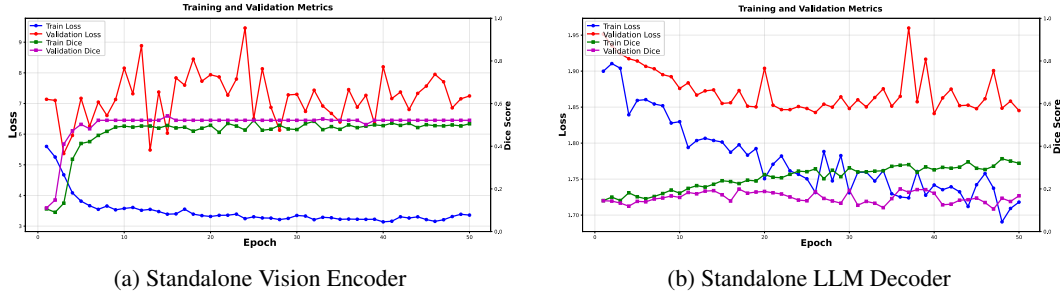


Figure 3: Impact of architecture design. Left: model trained exclusively on vision encoder embeddings. Right: model trained exclusively on LLM decoder embeddings. Both experiments demonstrate that neither modality alone achieves performance comparable to the proposed combined approach.

By combining the embeddings from our newly adapted components—the Vision Encoder and the LLM head—the model obtains a rich joint representation. Both embedding sets are projected into a shared latent space, fused via self-attention (Proj + Att in Fig. 2), and subsequently decoded by dedicated heads. While Figure 2 and the preceding discussion capture the core architectural design, we refer interested readers to Appendix B.2 for a closer look at the internal functioning.

6 Experimental Protocol

6.1 AIdocFactory Generated Data

While AIdocFactory processes diverse input formats (e.g., DOCX, PPTX, HTML), this experimental section focuses on PDFs which constitute the vast majority of online documents [2, 10]. We source our real documents from arXiv, an open repository that allows massive data collection without strict privacy constraints. arXiv offers two critical advantages. First, its categorization by scientific field (e.g., Computer Science, Biology, Economics) ensures high variability. Second, it allows restriction to peer-reviewed conference and journal articles, which serves as a quality control mechanism.

Using arXiv’s temporal metadata, we select 80% of our articles from before 2021, ensuring the dataset largely predates widespread LLM usage. However, because document patterns (e.g., layouts, plotting conventions, writing styles) evolve over time [6], the remaining 20% is sampled from post-2021 publications. While introducing a marginal risk of mislabeling some AI-generated content as real, we argue that capturing the natural evolution of modern authoring practices outweighs this minimal potential inclusion. Additionally, our strict peer-review filtering further mitigates this risk.

Using this curation strategy, we collect 2,000 real documents and generate 3,000 AI-modified counterparts via AIdocFactory, resulting in a robust dataset of over 35,000 pages. Crucially, *all* documents—including the pristine, unmodified real set—are processed and rendered through the AIdocFactory export pipeline. That is, not only are real and AI-generated components within a modified document exported identically (as explained in Sec. 4), but completely unmodified real documents also undergo the exact same rendering process. While AIdocFactory is explicitly designed to avoid low-level generative artifacts, this exhaustive measure acts as a failsafe. It guarantees that it is impossible to rely on export discrepancies. Consequently, the sole distinguishing factor remains the intrinsic semantic and structural nature of the content itself, forcing the model to learn meaningful forensic representations—a claim further supported by Sec. 6.5 and the analyses in A.2 and A.3.

A primary challenge in AIGC detection across modalities is generalizing to new generative models [44, 60]. Detectors typically overfit to the specific generation patterns of their training data, consequently failing when evaluated on content from other generators. For this reason, it is critical to evaluate performance not only on data produced by the same training generators (In-generation), but also on data from completely unseen models (Cross-Generation). Further details regarding the employed generative models and specific AIdocFactory hyperparameters are provided in Appendix A.5.

Retraction Watch Benchmark As highlighted in Sec. 4, the absolute lack of existing data necessitated the development of AIdocFactory. During our literature review, the only available proxy we identified was Retraction Watch [46], which catalogs scientific papers withdrawn from peer-reviewed venues primarily due to glaring LLM-generated artifacts (e.g., forgotten “Regenerate Response” prompts). However, this data suffers from severe limitations: (1) extreme scarcity; (2) reliance on explicit slip-ups rather than intrinsic generative patterns, fundamentally altering the detection task; (3) a lack of verified and precise ground-truth; and (4) an exclusive focus on text generation.

Despite these flaws, an extensive curation of this repository yielded 53 viable examples for an exploratory evaluation. The sole objective is to observe whether detection trends align with our benchmark. We strongly emphasize that due to its severe limitations, performance on this dataset serves strictly as an observational reference. The definitive metric of our model’s capability remains comprehensively established by the AIdocFactory benchmark.

6.2 AIdocDetector Model Training

AIdocDetector operates on document pages. The training set comprises 15,000 pages derived from Sec. 6.1 In-Generation data, maintaining a class distribution of approximately 66% AI-generated and 33% pristine pages. For evaluation, both the In-Generation and Cross-Generation test sets contain 10,000 pages each, strictly balanced with a 50/50 split of real and AI-generated samples.

AIdocDetector was trained on a NVIDIA H100 (94GB) GPU for approximately 15 days. To maximize memory efficiency and computational throughput, the entire pipeline was executed using `bf16` mixed precision. We utilized a batch size of 12. Specific hyperparameters are provided in Appendix B.1. The training pipeline is structured into two distinct phases. Phase 1 optimizes the entire architecture—excluding the classification head. Phase 2 freezes these learned weights and trains only the classification head for a few epochs. Because the dense embeddings learned during phase 1 already encode the presence or absence of AIGC (as 33% of the samples are pristine), they inherently capture the necessary classification signals. Consequently, Phase 2 serves as a rapid alignment step to map these rich embeddings to binary logits, avoiding the need for arbitrary heuristic thresholds. Learning curves for both phases are provided in Appendix B.1.

6.2.1 Loss Function

We employ different loss functions depending on the training phase. Phase 1 (segmentation) uses a combined focal and Dice loss following the SAM framework:

$$\mathcal{L}_{\text{total}} = 2.0\mathcal{L}_{\text{focal}} + 1.0\mathcal{L}_{\text{Dice}} \quad (1)$$

where focal loss is

$$\mathcal{L}_{\text{focal}} = \alpha(1 - p_t)^\gamma \text{BCE}(z, y) \quad (2)$$

($\alpha = 0.5$, $\gamma = 2.0$), BCE denotes binary cross-entropy with logits, z are the raw model predictions, $y \in \{0, 1\}$ are the ground truth binary masks, and p_t is the predicted probability of the true class. Dice loss measures prediction-ground truth overlap ($\epsilon = 0.01$):

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum p_i y_i + \epsilon}{\sum p_i + \sum y_i + \epsilon} \quad (3)$$

Phase 2 (classification) uses weighted cross-entropy $\mathcal{L}_{\text{CE}}$ with weights $w_1 = 0.66$, $w_2 = 0.33$:

$$\mathcal{L}_{\text{CE}} = -(w_1 y_1 \log(p_1) + w_2 y_2 \log(p_2)) \quad (4)$$

Test set	Model	Segmentation				Classification				
		Dice	IoU	Sensitivity	Specificity	F1	AUROC	Acc	Precision	Recall
In-Generation	Baseline	✗	✗	✗	✗	0.621	—	0.618	0.641	0.603
	GPT-5.2	✗	✗	✗	✗	0.411	0.621	0.581	0.640	0.303
	Gemini-3-flash	✗	✗	✗	✗	0.763	0.821	0.794	0.876	0.676
	Claude-opus-4.5	✗	✗	✗	✗	0.272	0.676	0.582	0.875	0.161
	Ours	0.880	0.854	0.835	0.990	0.833	0.867	0.839	0.833	0.833
Cross-Generation	Baseline	✗	✗	✗	✗	0.612	—	0.617	0.630	0.595
	GPT-5.2	✗	✗	✗	✗	0.428	0.612	0.570	0.639	0.294
	Gemini-3-flash	✗	✗	✗	✗	0.777	0.823	0.800	0.877	0.697
	Claude-opus-4.5	✗	✗	✗	✗	0.263	0.664	0.570	0.917	0.148
	Ours	0.847	0.817	0.711	0.996	0.827	0.846	0.828	0.831	0.823
Retraction Benchmark	Baseline	✗	✗	✗	✗	0.322	—	0.582	0.661	0.213
	GPT-5.2	✗	✗	✗	✗	0.295	0.695	0.590	1.000	0.173
	Gemini-3-flash	✗	✗	✗	✗	0.450	0.712	0.628	0.842	0.307
	Claude-opus-4.5	✗	✗	✗	✗	0.447	0.752	0.647	1.000	0.288
	Ours	*	*	*	*	0.602	0.810	0.692	0.806	0.481

Table 1: Performance comparison on the In-Generation, Cross-Generation, and Retraction datasets. Our model achieves robust segmentation (Dice: 0.880/0.847) with high specificity (>0.99) and superior classification (F1: 0.833/0.827/0.602). It consistently outperforms MLLM baselines across all datasets while maintaining a balanced precision-recall tradeoff. ✗: No segmentation capability; —: Lacks classification probabilities (no AUROC); *: No pixel-level ground truth.

6.3 In-Generation and Cross-Generation evaluation

Table 1 compares AidocDetector’s segmentation and classification performance against baselines across three test sets. A key finding is its robust generalization to content from unseen generative models (Cross-Generation), yielding nearly identical classification and highly consistent segmentation results versus the In-Generation setting. The sole exception is a drop in sensitivity, primarily driven by a threshold mismatch, a phenomenon that must be understood through the sensitivity-specificity trade-off. As illustrated in Appendix D.1, the decision threshold can be recalibrated to navigate this trade-off. Although this balance could be further adjusted during training—for instance, by tuning the Focal Loss α (Eq. 2) or incorporating Dice Loss class weights (Eq. 3) to penalize false negatives—we deliberately favor high specificity. In AIGC detection, minimizing false positives is paramount to avoid unjust accusations; therefore, this trade-off was explicitly chosen to reflect this priority.

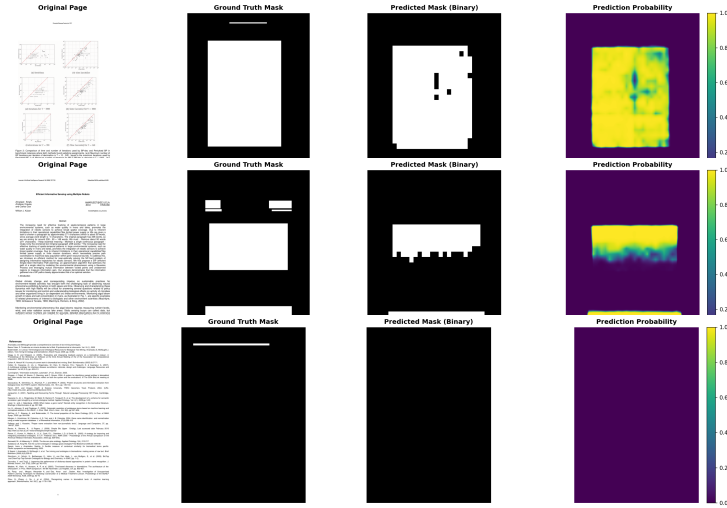


Figure 4: Qualitative analysis of AidocDetector. First and second examples illustrate segmentation challenges: the model struggles with boundary precision and spatial continuity. All examples show a loss in the localization of very small modifications.

6.4 Benchmarking against Frontier MLLMs

As the first to address document-level AIGC detection, we establish a rigorous benchmark by comparing our approach against state-of-the-art Multimodal Large Language Models (MLLMs). These

models consistently demonstrates frontier performance across a broad spectrum of tasks, frequently rivaling or surpassing specialized systems [53, 68, 5]. Indeed, a growing body of literature focuses on evaluating these models, often yielding remarkable results that redefine baseline expectations [35]. Consequently, benchmarking against these MLLMs is a necessary step to justify the development of specialized architectures in the current AI landscape. Additionally, we include a zero-shot baseline derived from [23], which we adapted for document-level AIGC detection and which served as our initial exploratory approach; the code for this adaptation is also publicly released². This approach is predicated on the hypothesis that intermediate model representations for genuine documents exhibit greater stability under input perturbations than those for AIGC.

Table 1 presents the comparative results. A primary finding—and a distinct advantage of our dedicated system—is that even these leading MLLMs fail to generate pixel-level outputs for document segmentation. This limitation restricts direct comparison to the classification task, where our model consistently outperforms all baselines across the three test sets. A per-class analysis reveals a critical behavioral divergence: while MLLMs exhibit a pronounced bias toward “Real” classifications—characterized by high precision but exceptionally low recall, particularly for Claude and GPT-4—our approach maintains an optimal balance, effectively detecting both genuine and AIG documents. This robustness persists in the Retraction Benchmark; while performance naturally favors the “Real” class, our model still achieves 0.481 recall and 0.806 precision for AIG documents.

6.5 Qualitative Results

To gain deeper insights beyond numerical metrics, we qualitatively analyze the test set results. Figure 4 illustrates representative examples, with additional cases in Appendix D and our GitHub¹.

Qualitative analysis shows that most detection failures involve minor modifications (e.g., short text spans or isolated minor edits). This finding aligns closely with established literature in AIGC detection, which demonstrates that in small-scale inputs, insufficient generative patterns exist to reliably distinguish synthetic from authentic content [17, 41, 54]. Since our architecture is explicitly optimized to minimize false positives, it rarely flags these minor modifications. Nevertheless, we consider this conservative behavior to be methodologically rigorous: since the statistical distance between human and AI distributions diminishes in small-scale inputs, flagging these regions would likely indicate an over-reliance on spurious correlations rather than genuine generative signals. Furthermore, this inability to detect minor edits provides compelling empirical confirmation that AIdocFactory does not inadvertently introduce low-level export artifacts or structural fingerprints—if such trivial cues existed, the model would easily exploit them to detect even the smallest modifications.

Conversely, the model consistently identifies larger modifications. In these instances, imperfections are primarily limited to boundary imprecisions or incomplete spatial coverage. Because our primary objective is the robust localization of modified regions, these minor alignment deviations do not compromise the model’s forensic utility. The outputs remain highly actionable, providing a clear spatial prior that could be refined via standard post-processing if pixel-perfect precision were required.

7 Conclusions and Limitations

Addressing the unexplored challenge of AI-generated document detection, we presented AIdoc: a framework integrating AIdocFactory, a modular synthetic generation pipeline, and AIdocDetector, a novel VLM-based forensic architecture. Evaluations demonstrate that AIdocDetector significantly outperforms MLLMs and exhibits strong In- and Cross-Generation performance. Together, these components provide a scalable and robust solution for preserving document integrity.

Several limitations should be noted. First, detecting increasingly smaller modifications remains inherently difficult due to insufficient statistical signal in short sequences. Future work should aim to progressively refine the detection of these minor modifications, pushing it to the theoretical bounds of AIGC detection. Second, exhaustive hyperparameter optimization would be valuable to determine the maximum performance ceiling achievable by our proposed approach. Finally, a large-scale “in-the-wild” evaluation—similar to Retraction Watch, but utilizing a massive, flawlessly annotated corpus, free of labeling “slip-ups”, and covering diverse modification types—would be highly insightful. While our dataset adequately proxies this evaluation, such in-the-wild validation remains an important milestone once these data become available.

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552 A AIdocFactory.

553 A.1 Database Schema

554 The following shows an example of what is stored in the database automatically generated by
555 AIdocFactory.

- 556 • **pdf_path**: Full path to the original PDF file (string).
- 557 • **is_modified**: Boolean indicating whether the document has been artificially modified
558 (true/false).
- 559 • **modifications**: Array of objects specifying the modified regions when `is_modified =`
560 `true`. Each modification includes:
 - 561 – **component_type**: Type of affected component (“section_header”, “footnote”,
562 etc.).
 - 563 – **type**: Semantic subtype (“text”, “image”).
 - 564 – **page**: Affected page number (int).
 - 565 – **bbox**: Bounding box of the modified region, with structure:
 - 566 * **page, x, y**: Coordinates and page.
 - 567 * **width, height**: Bounding box dimensions.
 - 568 * **coord_origin**: Coordinate system origin (e.g., “CoordOrigin.BOTTOMLEFT”).
 - 569 * **raw**: Original layout coordinates (left, top, right, bottom).
- 570 • **component_inventory**: Complete structural inventory of the PDF, containing:
 - 571 – **pages**: Dictionary indexed by page number (“1”, “2”, ...), where each page contains:
 - 572 * **page_number**: Page number (int).
 - 573 * **text_sections**: Array of text blocks, each with:
 - 574 · **type**: “TextItem”, “SectionHeaderItem”, etc.
 - 575 · **label**: Semantic label (“text”, “section_header”, “page_header”, etc.).
 - 576 · **parent_type**: Parent container (“body”, “picture”, “table”).
 - 577 · **preview**: Representative text snippet.
 - 578 · **length**: Character length.
 - 579 · **bounding_box**: Same structure as `modifications.bbox`.
 - 580 · **ref**: Internal reference (“#/texts/35”).
 - 581 * **tables_in_page**: Array of detected tables.
 - 582 * **images_in_page**: Array of detected images.

583 **A.2 AIdocFactory Qualitative Comparison**

584 Figure 5 presents three examples illustrating the functionality of AIdocFactory, with each demonstrat-
 585 ing different modifications. As observed, the sole difference between the real and modified samples
 586 is the content itself. This ensures the model is forced to detect intrinsic generative patterns rather
 587 than relying on visual artifacts. Additional examples are available in our GitHub repository ¹.

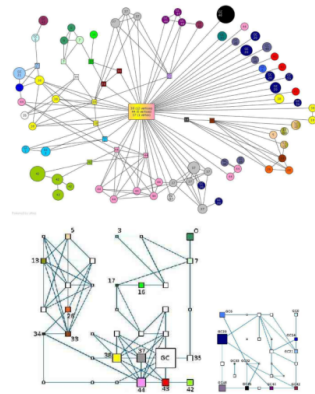


Fig. 9. Comparison of the representation of the graph through perfect communities and self-organizing maps (left : of the whole graph; right : of the largest cluster of the initial SOM. More details about this figure are given in the text)

SOM: this is the case, for example, for clusters 44 and 37, for clusters 0 and 7, for clusters 26, 33 and 34, etc. It is also interesting to note that some of the persons having a high betweenness measure also have an important position on the SOM: for example, peasants 17 and 34 are emphasized by the fact that they link the bottom right part of the map with the top right part and the

26

(a) Original Page

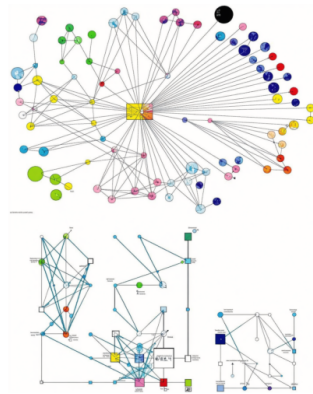


Fig. 9. Comparison of the representation of the graph through perfect communities and self-organizing maps (left : of the whole graph; right : of the largest cluster of the initial SOM. More details about this figure are given in the text)

SOM: this is the case, for example, for clusters 44 and 37, for clusters 0 and 7, for clusters 26, 33 and 34, etc. It is also interesting to note that some of the persons having a high betweenness measure also have an important position on the SOM: for example, peasants 17 and 34 are emphasized by the fact that they link the bottom right part of the map with the top right part and the

26

(b) Original Page With Image Modifications

Figure 5: AIdocFactory examples. We show three cases demonstrating our system’s functionality: (1) primarily text modification (continued on next page), (2) primarily table modification, and (3) primarily image modification. Each row displays the original document component (left) and its AI-modified version (right).

In the absence of such knowledge, I therefore had to choose the equally possible alternative of identifying such anti-peptide string by searching for proteins/peptides known to antagonize/attenuate pathologic aspects of RA. My search yielded calcitonin as a possible candidate given that this hormone has been shown to have beneficial effects in the treatment of RA (19,20) which may be due to its previously demonstrated ability to inhibit macrophage function (21). Under the premise that calcitonin may contain peptide information overlapping with an existing anti-arthritis peptide string that involves RB and CHC under physiological circumstances, I performed a sequence comparison of all these molecules in the form of tripeptide sequences. This analysis yielded that RB and calcitonin share only the 2 tripeptides Gly-Asn-Leu (GNL) and Asn-Leu-Ser (NLS) whereas CHC and calcitonin share the 7 tripeptides GNL, LST, TOD, QTA, TAI, IGV and VGA.

Moreover, to achieve a perfect functional mirroring to the TKVD-based peptide string, an immunoglobulin had to be searched for that possibly shares with RB and/or CHC as well as calcitonin one of the above tripeptides. Following this insight, my additional exploration has yielded that, remarkably, the variable region of a distinct human immunoglobulin A1 heavy chain, briefly IgA1 VH (22), bears the NLS tripeptide whereas a comparable domain within the human IgG1 heavy chain (23) does not contain any of the above tripeptides. Notably, CHC harbors (in either of its two isoforms) the highly related sequence N F S. Along the same lines, I have detected the NLS sequence also in CD16 (i.e. a receptor specifically recognizing the constant region of its ligand IgG, briefly an Fc receptor), thus supporting the possibility of a direct physical interaction between TKVD (n.e.g. in IgG) and NLS (e.g. in CD16) peptide sequences, besides their above anticipated involvement in mediating opposite biological effects.

Since such found NLS sequences conform with the universal glycosylation motif NXS whereby X signifies any amino acid, the following intriguing scenario is conceivable. Normally, the NLS sequence in IgA1 VH may fulfill 2 roles:

- to antagonize Candida infection by neutralizing the TKVD peptide in the Rac1 protein of this fungus;
- to mask the TKVD epitidotope in IgG1 such that it is not recognized as an auto-antigenic site. As far as this second point is concerned and expressed differently, I assume that physiologically there are some IgA1 anti-idiotypic antibodies that, as part of an anti-arthritis idiotype network, suppress the "surfacing" of potentially autoimmune sites (in the constant region of IgG1. Such host-beneficial masking of this IgG1 site would probably be performed not only by IgA1, but also by CD16. In this context, it should be specified that the TKVD tetrapeptide is located in close proximity to the IgG hinge region whose conformational change presumably accounts for the structurally altered serum IgG of RA patients as compared to that of healthy individuals (24,25), thus supporting the possibility that IgG1 TKVD may be unmasked as an auto-epitope for RF. Consistent with this prediction, Johnson and coworkers had precisely supposed that "new determinants are exposed at or near the hinge region of the rheumatoid IgG molecule" (26). Notably, the investigations carried out by Johnson et al. (25,26) have remained particularly important since they were performed with RA patient-derived IgG whereas subsequent crystallization or

Given the lack of specific knowledge about the anti-peptide string, I selected the plausible alternative of identifying it by searching for proteins or peptides that counteract the pathological aspects of rheumatoid arthritis (RA). My research led to calcitonin as a potential candidate, as this hormone has demonstrated therapeutic benefits in treating RA, attributed to its ability to inhibit macrophage function. I conducted a sequence comparison of RB, CHC, and calcitonin using tripeptide sequences. This analysis revealed that RB and calcitonin share only two tripeptides, Gly-Asn-Leu and Asn-Leu-Ser, whereas CHC and calcitonin share seven tripeptides, including Gly-Asn-Leu and others. Note: This analysis supports the assumption that calcitonin may contain overlapping peptide information with an existing anti-arthritis peptide string involving RB and CHC under physiological conditions.

A specific immunoglobulin had to be identified that shares similarities with RB and CHC, as well as calcitonin, in terms of certain tripeptides. My investigation revealed a unique human immunoglobulin A1 heavy chain variant, IgA1 VH (22), that exhibits the NLS tripeptide, whereas a comparable domain in the human IgG1 heavy chain (23) does not contain any of these tripeptides. CHC's two isoforms both possess the sequence N F S, which is highly similar to the NLS tripeptide. Note: Minor adjustments to wording were made to improve clarity and flow while preserving the original meaning and context. The rewritten paragraph is within 5-10% variation from the original. Similarly, I have identified the NLS sequence in CD16, a receptor that specifically recognizes the constant region of its ligand, IgG, which is also referred to as an Fc receptor. This finding supports the notion of a direct physical interaction between the TKVD peptide sequences found in IgG and NLS sequences found in CD16, in addition to their anticipated roles in mediating opposite biological effects.

Since such found NLS sequences conform with the universal glycosylation motif NXS whereby X signifies any amino acid, the following intriguing scenario is conceivable. Normally, the NLS sequence in IgA1 VH may fulfill 2 roles:

- To counteract the Candida infection, scientists have developed a method that targets the Rac1 protein of this fungus.
- To counteract the TKVD epitidotope in IgG1, the immune system employs a mechanism where IgA1 anti-idiotypic antibodies mask the constant region of IgG1, preventing it from being recognized as an auto-antigenic site. CD16 likely plays a role in suppressing the "surfacing" of potentially autoimmune sites in IgG1, facilitating this process. Consequently, this host-beneficial masking of the IgG1 site prevents the immune system from responding to it as an auto-antigen.

In this context, it should be specified that the TKVD tetrapeptide is located in close proximity to the IgG hinge region whose conformational change presumably accounts for the structurally altered serum IgG of RA patients as compared to that of healthy individuals (24,25), thus supporting the possibility that IgG1 TKVD may be unmasked as an auto-epitope for RF. Consistent with this prediction, Johnson and coworkers had precisely supposed that "new determinants are exposed at or near the hinge region of the rheumatoid IgG molecule" (26). Notably, the investigations carried out by Johnson et al. (25,26) have remained particularly important since they were performed with RA patient-derived IgG whereas subsequent crystallization or

(c) Original Page

RefSeq	Rheumatoid arthritis peptide strings	arXiv August 10, 2018
Tab. 1	Arthritis peptide string	Anti-arthritis peptide string
Extracellular environment	Thr-Lys-Val-Asp-hum-IgG1 C H (aa 209-212)	Asn-Leu-Ser Asn-Leu-Ser Asn-Leu-Ser
Cytoplasm	Thr-Lys-Val-Asp-hum-IgG1 C H (aa 209-212) Thr-Lys-Val-Asp-hum-CHC1 (aa 1608-1611) Thr-Lys-Val-Asp-hum-RB (aa 142-145)	Asn-Leu-Ser Phe-Ser Asn-Leu-Ser
Nucleus	Thr-Lys-Val-Asp-hum-CHC1 (aa 1608-1611) Thr-Lys-Val-Asp-hum-RB (aa 142-145)	Asn-Phe-Ser Asn-Leu-Ser
Tab. 2	Arthritis peptide string	Anti-arthritis peptide string
Extracellular environment	Thr-Lys-Val-Asp-Calcitonin Receptor 1 (aa 116-119) Thr-Lys-Val-Asp-hum-IgG1 C H (aa 209-212)	Asn-Leu-Ser Asn-Leu-Ser Asn-Leu-Ser
Cytoplasm	Thr-Lys-Val-Asp-Calcitonin Receptor 1 (aa 116-119) Thr-Lys-Val-Asp-hum-IgG1 C H (aa 209-212) Thr-Lys-Val-Asp-hum-CHC1 (aa 1608-1611) Thr-Lys-Val-Asp-hum-RB (aa 142-145)	Asn-Leu-Ser Phe-Ser Asn-Leu-Ser
Nucleus	Thr-Lys-Val-Asp-hum-CHC1 (aa 1608-1611) Thr-Lys-Val-Asp-hum-RB (aa 142-145)	Asn-Phe-Ser Asn-Leu-Ser
Tab. 3	Arthritis peptide string	Anti-arthritis peptide string
Extracellular environment	[Thr-Lys-Val-Asp-Calcitonin Receptor 1 (aa 116-119) Thr-Lys-Val-Asp-hum-IgG1 C H (aa 209-212)]	
Cytoplasm	[Thr-Lys-Val-Asp-Calcitonin Receptor 1 (aa 116-119) Thr-Lys-Val-Asp-hum-IgG1 C H (aa 209-212) Thr-Lys-Val-Asp-hum-CHC1 (aa 1608-1611) Thr-Lys-Val-Asp-hum-RB (aa 142-145)]	Asn-Phe-Ser Asn-Leu-Ser Asn-Leu-Ser
Nucleus	Thr-Lys-Val-Asp-hum-CHC1 (aa 1608-1611) Thr-Lys-Val-Asp-hum-RB (aa 142-145)	Asn-Phe-Ser Asn-Leu-Ser

(e) Original Page

(d) Original Page With Text Modifications

RefSeq	Arthritis peptides in autoimmune region	arXiv August 10, 2018
Tab. 1	Arthritis peptide string	Anti-arthritis peptide string
Extracellular environment	Thr-Lys-Val-Asp-hum-IgG1 C H (aa 209-212)	Asn-Leu-Ser Asn-Leu-Ser Asn-Leu-Ser
Cytoplasm	Thr-Lys-Val-Asp-hum-IgG1 C H (aa 209-212) Thr-Lys-Val-Asp-hum-CHC1 (aa 1608-1611) Thr-Lys-Val-Asp-hum-RB (aa 142-145)	Asn-Leu-Ser Phe-Ser Asn-Leu-Ser
Nucleus	Thr-Lys-Val-Asp-hum-CHC1 (aa 1608-1611) Thr-Lys-Val-Asp-hum-RB (aa 142-145)	Asn-Phe-Ser Asn-Leu-Ser
Tab. 2	Arthritis peptide string	Anti-arthritis peptide string
Anti-arthritis peptide string 1	Thr-Lys-Val-Asp-Calcitonin Receptor 1 (aa 116-119) Thr-Lys-Val-Asp-hum-IgG1 C H (aa 209-212)	Asn-Leu-Ser Asn-Leu-Ser Asn-Leu-Ser
Anti-arthritis peptide string 2	Thr-Lys-Val-Asp-hum-CHC1 (aa 1608-1611) Thr-Lys-Val-Asp-hum-RB (aa 142-145)	Asn-Phe-Ser Asn-Leu-Ser
Tab. 3	Arthritis peptide string	Anti-arthritis peptide string
Extracellular environment	[Thr-Lys-Val-Asp-Calcitonin Receptor 1 (aa 116-119) Thr-Lys-Val-Asp-hum-IgG1 C H (aa 209-212)]	
Cytoplasm	[Thr-Lys-Val-Asp-Calcitonin Receptor 1 (aa 116-119) Thr-Lys-Val-Asp-hum-IgG1 C H (aa 209-212) Thr-Lys-Val-Asp-hum-CHC1 (aa 1608-1611) Thr-Lys-Val-Asp-hum-RB (aa 142-145)]	Asn-Phe-Ser Asn-Leu-Ser Asn-Leu-Ser
Nucleus	Thr-Lys-Val-Asp-hum-CHC1 (aa 1608-1611) Thr-Lys-Val-Asp-hum-RB (aa 142-145)	Asn-Phe-Ser Asn-Leu-Ser

(f) Original Page with Table Modifications

A.3 AIdocFactory Quantitative Analysis

To complement the qualitative evaluation of our framework, we conduct a rigorous quantitative analysis to evaluate the structural, visual, and statistical fidelity of the AIG documents compared to their real counterparts, we implemented four distinct forensic tests:

- **Frequency Analysis (FFT):** We evaluate the Fast Fourier Transform (FFT) spectra to identify synthetic frequency fingerprints. Generative models often leave characteristic traces in high-frequency bands; thus, we calculate total energy and the distribution ratios across low, mid, and high frequencies.
- **OCR Confidence and Lexical Integrity:** To evaluate character-level distortions, legibility, and text degradation, we process the documents through a standard OCR engine. We compare mean confidence scores, total word counts, Out-Of-Vocabulary (OOV) rates, and gibberish ratios.
- **Layout Geometry:** Synthetic rendering can sometimes produce “hyper-regular” alignments or abnormal spacing. We measure the standard deviation of left and right margins, as well as the Coefficient of Variation (CV) of word gaps.
- **Error Level Analysis (ELA):** To isolate non-uniform compression signatures typical of manipulated images, we apply ELA, analyzing the mean and standard deviation of compression errors across text regions and background areas.

Results and Discussion. We tested 2,500 AIG pages generated by AIdocFactory against 2,500 pristine real pages. The comparative results are presented in Table 2. As observed, the numerical differences across all four forensic domains are negligible. The minor fluctuations present are purely stochastic, arising from the natural variability inherent to distinct document layouts and contents, rather than their source (real vs. AIG). This extremely high statistical overlap demonstrates that AIdocFactory successfully replicates the physical and digital properties of real-world documents. Consequently, this validates our methodological claim: detection models trained on this benchmark are forced to learn meaningful, high-level generative patterns, as low-level structural, frequency, or compression shortcuts are absent. Moreover, we emphasize that this behavior is also evident in the results presented in Sec. 6.5.

Table 2: Quantitative evaluation of low-level artifacts between AIG and real documents across 5,000 samples. Metrics indicate near-perfect statistical alignment across frequency, legibility, geometry, and compression domains.

Metric	Fake (AIG)	Real
Frequency Domain (FFT)		
Total energy	$(2.19 \pm 0.11) \times 10^{11}$	$(2.19 \pm 0.11) \times 10^{11}$
Low ratio	0.980 ± 0.009	0.980 ± 0.009
Mid ratio	0.007 ± 0.004	0.007 ± 0.004
High ratio	0.013 ± 0.006	0.013 ± 0.006
OCR Integrity & Confidence		
Mean conf.	49.43 ± 16.64	49.90 ± 16.41
Word count	170.6 ± 84.7	170.5 ± 83.8
OOV rate	0.486 ± 0.192	0.466 ± 0.190
Gibberish rate	0.023 ± 0.024	0.024 ± 0.023
Layout Geometry		
L-margin std	123.5 ± 18.3	123.6 ± 18.4
R-margin std	125.3 ± 17.7	125.3 ± 18.2
Word gap CV	1.386 ± 0.299	1.352 ± 0.290
Error Level Analysis (ELA)		
Mean	0.735 ± 0.261	0.728 ± 0.268
Std	1.749 ± 0.425	1.743 ± 0.431
Text mean	3.123 ± 0.737	3.108 ± 0.690
Bg. mean	0.541 ± 0.191	0.536 ± 0.195

616 A.4 AIdocFactory: Docling Structured Variable

617 DocLing processes PDF documents and returns a `ConversionResult` object containing all struc-
618 tured information about the document. Below we describe its hierarchical structure:

619 A.4.1 Level 1: `ConversionResult`

620 The root structure contains global metadata and data collections:

```
621 ConversionResult(  
622     input: InputDocument,           # Input PDF information  
623     document_hash: str,             # Unique document hash  
624     status: ConversionStatus,       # SUCCESS / FAILURE / ...  
625     errors: List[str],              # Errors during conversion  
626     pages: List[Page],              # Processed pages  
627     assembled: AssembledUnits       # Unified elements  
628 )  
629  
630
```

631 A.4.2 Level 2: `InputDocument`

632 Contains metadata about the input file:

```
633 InputDocument(  
634     file: PurePosixPath,            # PDF file path  
635     format: InputFormat,            # PDF, DOCX, etc.  
636     filesize: int,                  # Size in bytes  
637     pagecount: int,                 # Total pages (23)  
638     document_hash: str,             # Document hash  
639     valid: bool                     # Document validity  
640 )  
641  
642
```

643 A.4.3 Level 3: `Page`

644 Each page contains size information and layout predictions:

```
645 Page(  
646     pageno: int,                    # Page number (0-indexed)  
647     size: Size,                     # width=612.0, height=792.0 (points)  
648     predictions: PagePredictions    # Element predictions  
649 )  
650  
651
```

652 A.4.4 Level 4: `PagePredictions (LayoutPrediction)`

653 Contains clusters of detected elements on the page:

```
654 PagePredictions(  
655     layout: LayoutPrediction(  
656         clusters: List[Cluster],    # Elements: texts, headers, images  
657         tablemap: dict,             # Table mapping (if exists)  
658         figures: dict               # Figure detection (None)  
659     )  
660 )  
661  
662
```

663 A.4.5 Level 5: Cluster (Content Element)

664 Each cluster represents an element (paragraph, heading, image, etc.):

```
665 Cluster(  
666     id: str,                                # Unique ID (e.g., "13", "5")  
667     label: DocItemLabel,                    # TEXT | SECTIONHEADER | PICTURE |  
668     ...  
669     bbox: BoundingBox,                      # Element bounding box  
670     confidence: float,                      # Confidence: 0.51-0.99  
671     cells: List[PdfTextCell],               # Text cells within cluster  
672     children: List[Cluster]                 # Nested clusters  
673 )  
674  
675
```

676 A.4.6 Level 6: BoundingBox

677 Defines element coordinates on the page:

```
678 BoundingBox(  
679     l: float,                                # left (77.4)  
680     t: float,                                # top (38.687)  
681     r: float,                                # right (277.637)  
682     b: float,                                # bottom (93.063)  
683     coord_origin: CoordOrigin               # TOPLEFT  
684 )  
685  
686
```

687 A.4.7 Level 7: PdfTextCell

688 Represents each text cell within a cluster:

```
689 PdfTextCell(  
690     index: int,                              # Index in cluster (0, 1, 2, ...)  
691     text: str,                               # "Transactions of the SDPS"  
692     orig: str,                               # Original text  
693     confidence: float,                       # 1.0 (very confident)  
694     textdirection: TextDirection,           # LEFTTORIGHT  
695     font_key: str,                           # "T10" (font reference)  
696     font_name: str,                         # "TimesNewRomanPSMT"  
697     rendering_mode: RenderingMode,          # UNKNOWN, VISIBLE, etc.  
698     rect: BoundingBox,                      # Specific cell coordinates  
699     rgba_color: RGBAColor                  # r=0, g=0, b=0, a=255  
700 )  
701  
702
```

703 A.4.8 Level 8: AssembledUnits (Unified Elements)

704 Collects all processed elements in a unified structure:

```
705 AssembledUnits(  
706     elements: List[Union[TextElement, ImageElement]]  
707 )  
708  
709 TextElement(  
710     label: DocItemLabel,                    # Element type  
711     id: str,                                # Unique ID  
712     pageno: int,                            # Page number (0-indexed)  
713     cluster: Cluster,                       # Cluster reference  
714     text: str                               # Combined text  
715 )  
716  
717
```

A.5 AIdocFactory: Generation Hyperparameters

This appendix details all hyperparameters used in AIdocFactory for document generation. The main generation aspects are discussed in Section 6.1; here we provide complete technical specifications.

A.5.1 Generative Models

Generative models are broadly categorized into Language and Vision domains. The *Language* models include DialoGPT [67], Phi-3-mini-4k-instruct [1], Qwen2.5 [64], Qwen3 [65], Llama-3.2 [20], and Granite-3.0 [19], while the *Vision* models consist of Stable-Diffusion-v1.5 [47] and SD-Turbo [48]. For our **In-Generation** dataset (comprising training, validation, and testing splits), we employ the first four language models and the first vision model. Conversely, the **Cross-Generation** set (testing splits) utilizes the remaining models. Notably, our framework affords the flexibility to selectively assign specific vision or language models to distinct document components as required (see Sec. 4).

A.5.2 Similarity Level

Users control how closely the synthetic content of the components resembles the original through four predefined levels: very similar, similar, different, and completely different. While the similarity parameter is integrated into AIdocFactory to maximize its versatility for diverse downstream applications, we deliberately fix it to the “similar” setting for training AIdocDetector. We consider this the optimal configuration because it generates synthetic content that aligns closely with the original context, preventing trivial semantic discrepancies. Consequently, the model is forced to learn deep, intrinsic AI-generative patterns rather than potentially overfitting to thematic or contextual inconsistencies. Conversely, Under the “very similar” regime, the generative forensic signals are diluted, unnecessarily complicating the training process without contributing to robust AIGC identification. Therefore, the “similar” level provides the ideal balance for isolating the generative footprint during training.

A.5.3 Modification Ratio

This parameter governs the selection of document components targeted for modification. Initially, the framework parses the document to extract all underlying components—such as titles, paragraphs, lists, images, tables, footnotes, or formulas—structuring them hierarchically at both the global and page levels. Subsequently, target components can be selected either interactively or automatically via a stochastic ratio. Manual selection affords the user granular control to perform highly specific, targeted edits on individual documents. Conversely, for large-scale automated dataset generation, the stochastic ratio approach is preferred. This ratio defines the proportion of altered content: for instance, a ratio of 0.0 applies no changes, 0.5 alters half of the components, and 1.0 completely regenerates the document. To optimize the dataset for training the segmentation model, a ratio of 0.5 is highly recommended. It modifies half of the components at the document level—that means some pages might have just one element modified while others might have all elements changed. We designed it this way so our AIdocDetector learns across all types of contexts.

A.5.4 Text Generation Configuration

Temperature and sampling parameters are controlled by the similarity level (Table 3):

Similarity Level	Temperature	Top-p	Top-k	Rep. Penalty
Very Similar	0.5	0.95	80	1.0
Similar	0.7	0.90	60	1.02
Moderately Different	0.9	0.85	50	1.05
Totally Different	1.2	0.80	40	1.1

Table 3: Temperature and sampling parameters per similarity level for text generation.

Additional text generation constraints:

- **Minimum tokens:** 5
- **Maximum tokens:** 1024

- **Length calculation buffer:** 0.10 (10% margin)
- **LLM adjustment deviation threshold:** 0.20 (20%)
- **Max length adjustment iterations:** 2

762 A.5.5 Table Generation Configuration

763 Tables are generated with fixed sampling parameters:

- **Do sample:** True
- **Temperature:** 0.3
- **Max new tokens:** 3000

767 A.5.6 Image Generation Configuration

768 Image generation parameters vary by similarity level (Table 4):

Similarity Level	Strength	Guidance Scale	Inference Steps
Very Similar	0.25	6.0	10
Similar	0.45	7.5	10
Moderately Different	0.65	9.0	15
Totally Different	0.85	12.0	15

Table 4: Image generation parameters (image-to-image) per similarity level.

769 There are three image generation modes: image-to-image (generates from the replaced figure), text-
770 to-image (generates from the figure caption), or baseline (generates a random image if the previous
771 modes fail).

772 A.5.7 Document Processing Configuration

- **Image resolution scale:** 8.0
- **Generate page images:** True
- **Generate figure/picture images:** True
- **Export tolerance profiles:**
 - PDF Strict: 0.10 (10%)
 - PDF Basic: 0.20 (20%)

779 A.5.8 Hardware Configuration

- **Use GPU:** True
- **GPU ID:** 0
- **Device:** CUDA (fallback to CPU if unavailable)
- **GPU Memory:** 24 GB

784 B AIdocDetector

785 B.1 AIdocDetector: Training

786 Figure 6 presents the training curves for Phase 1, displaying loss curves (left) and validation metrics (right).
 787 (right). Figure 7 shows the training evolution of Phase 2, illustrating the progression of both loss
 788 values and performance metrics throughout the training process.

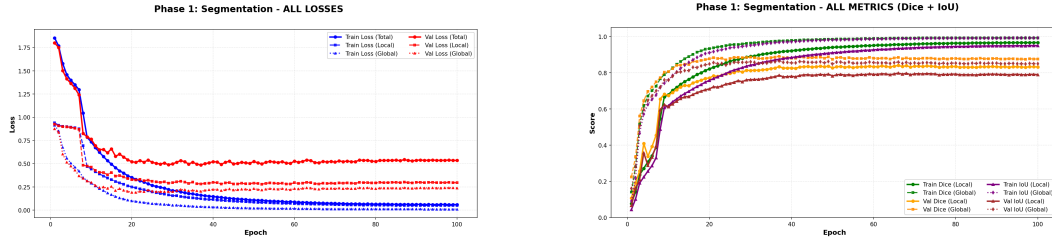


Figure 6: Phase 1 training curves: Loss progression (left) and validation metrics (right)

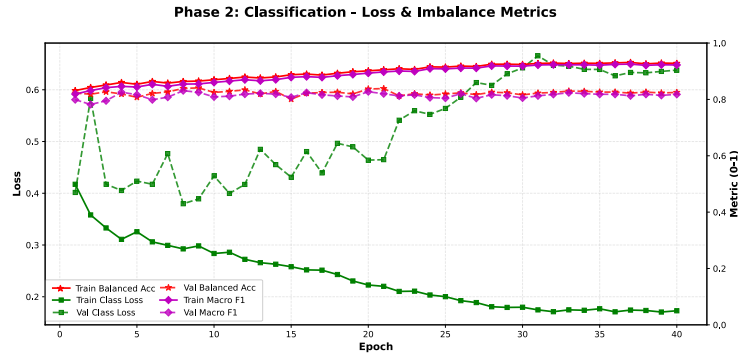


Figure 7: Phase 2 training curves showing loss and metrics evolution

789 B.1.1 Phase 1: Hyper-parameters

Param	Value
use_lora	true
r	16
lora_alpha	32
lora_dropout	0.1
bias	"none"
task_type	"Feature Extraction"

Table 5: LoRA configuration parameters

790 **LoRA Configuration** The target modules on which LoRA is applied are:

- 791 • q_proj (query projection)
- 792 • k_proj (key projection)
- 793 • v_proj (value projection)
- 794 • o_proj (output projection)
- 795 • gate_proj (gate projection)
- 796 • up_proj (up projection)
- 797 • down_proj (down projection)

798 **Training Protocol** Training is conducted for 100 epochs, but the checkpoint corresponding to the
799 epoch with the best performance on the validation set is selected.

800 **Optimizer and Scheduler** Phase 1 training uses the AdamW optimizer with the following parame-
801 ters:

Param	Value
Optimizer	AdamW
Initial learning rate	1.0×10^{-4}
Weight decay	0.05

Table 6: Phase 1 optimizer configuration

802 The learning rate is controlled through a sequential scheduler that combines a warmup phase and a
803 cooling phase:

- 804 • **Warmup:** LinearLR during the first 10% of training steps, scaling from 10^{-9} to the Initial
805 learning rate 10^{-4} .
- 806 • **Main scheduler:** CosineAnnealingLR with $\eta_{\min} = 10^{-6}$ for the remaining steps.

807 This approach improves training stability during initial iterations and enables smooth convergence
808 toward minimal learning rate values.

809 B.1.2 Phase 2: Hyper-parameters

810 Phase 2 training uses the same optimizer (AdamW) but with a higher learning rate of 1.0×10^{-3} ,
811 without additional scheduler. As discussed in the main text, 1-2 epochs are sufficient for this
812 classification head to align with the rich segmentation embeddings it builds upon. Nevertheless, we
813 trained for 40 epochs to demonstrate that training beyond 1-2 epochs is unnecessary.

814 The configurable parameters are set as follows:

- 815 • Number of epochs: 40 (for curve visualization)
- 816 • Learning rate: 1.0×10^{-3}
- 817 • Weight decay: 0.05
- 818 • Scheduler: None

819 B.2 AIdocDetector: Architecture Details

820 B.2.1 Recombiner

821 Before the Projection + Attention, the vision encoder embeddings are processed by an additional
822 recombination module (Recombiner module in Figure 2). The model input consists of the document
823 page processed as N image patches (local) plus the full page (global). Since embeddings are extracted
824 from four encoder depths, the Recombiner aggregates these $(N + 1) \times 4$ vectors into unified local
825 and global feature representations.

826 Figure 8 shows a more detailed diagram of the of the Recombiner module from Figure 2. It is a
827 simple module that, on one hand, separates the patch embeddings from the vision encoder and joins
828 them spatially in the same order in which they were obtained, so the rest of the model can analyze
829 them together with spatial coherence. On the other hand, it takes the global embedding of the entire
830 page. Additionally, it stacks the embedding blocks obtained at different depths, which represent
831 different levels of abstraction. This generates the representations that will be used by the rest of the
832 model.

833 B.2.2 Decoder Heads

834 After Projection + Attention, the model produces two multimodal embeddings—local and global.
835 These embeddings are then fed into the decoder heads. Because these embeddings already encode
836 highly informative and decoded features, the decoder heads can remain lightweight [62, 58]. Rather

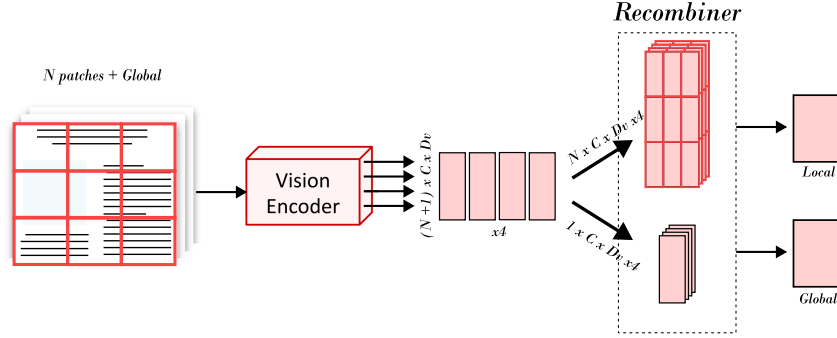


Figure 8: Internal architecture of the Recombiner module.

than relying on off-the-shelf decoders, we engineered custom architectural heads that strategically integrate core mechanisms from SegFormer [62], DETR [9], and FPN [37]. Their detailed designs are outlined below.

Segmentation Decoder Head: The architecture processes local and global features to generate binary segmentation masks. It employs independent projectors, spatial refinement, and an MLP-style FFN to enhance representations before final prediction.

The core components include:

- **Independent Projectors**, utilizing 1×1 convolutions with normalization (GroupNorm or BatchNorm) and ReLU to reduce input features into a common 256-dimensional embedding space.
- **Spatial Refinement** block, which employs depthwise-separable convolutions (3×3 depthwise and 1×1 pointwise) with skip connections for efficient local context capture.
- **MLP-style FFN** that refines the channel-wise representations via $2 \times$ expansion, GELU activation, and 1×1 convolution contraction.
- **Prediction Layer** uses a 1×1 convolution to generate segmentation logits, supported by optional auxiliary heads for multi-scale supervision.

Classification Decoder Head: The DecoderHeadClass is a simplified head designed to fuse local and global features for binary classification. This component performs spatial pooling and feature fusion to produce a global. Its architecture consists of:

- **Independent Projectors**, which use parallel 1×1 convolutions and normalization to map input features into a shared embedding space.
- **Attentional Pooling**, utilizing 1×1 convolutions to generate spatial attention weights that, via softmax, extract representative vectors (B , embed_dim) for each stream.
- a configurable **Fusion Mechanism** to combine the pooled vectors, offering three modes: *Cross-attention* (4-head multi-head attention where local features query global ones), *Gated* (an MLP-based gate for dynamic stream weighting), or *Weighted* (a linear combination with learnable softmax weights).
- **Classification Head** comprising LayerNorm, optional dropout, and a linear layer to transform the fused vector into classification logits.

866 **C Additional Qualitative Results**

867 In Figure 9, we can observe several other interesting examples in addition to the results presented in
 868 Section 6.5.



Figure 9: Qualitative analysis of AldocDetector performance, further examples.

D Additional Quantitative Results

D.1 Sensitivity-Specificity Trade-off and Threshold Analysis

The results presented in Table 1 should be interpreted within the context of the sensitivity-specificity trade-off. Our model has been explicitly trained to minimize false positives, a critical requirement in AI-generated content (AIGC) detection to avoid unjust accusations. However, maintaining such high specificity inevitably introduces a penalty on sensitivity.

To illustrate this dynamic, Table 7 demonstrates how the performance metrics shift simply by adjusting the decision threshold (in the $[0, 1]$ range) at inference time. Should a specific use case require a more balanced performance between these two metrics, the model’s operating point can be further adjusted during training by tuning the α parameter in the Focal Loss (Eq. 2) or by incorporating class weights into the Dice Loss (Eq. 3).

Threshold	Dice (In/Cross)	IoU (In/Cross)	Sensitivity (In/Cross)	Specificity (In/Cross)	F1 (In/Cross)
0.10	0.8864 / 0.8504	0.8595 / 0.8197	0.8624 / 0.7439	0.9871 / 0.9926	0.8723 / 0.7945
0.15	0.8855 / 0.8504	0.8586 / 0.8199	0.8565 / 0.7378	0.9878 / 0.9933	0.8692 / 0.7906
0.20	0.8850 / 0.8503	0.8581 / 0.8201	0.8522 / 0.7333	0.9883 / 0.9939	0.8669 / 0.7877
0.30	0.8832 / 0.8498	0.8564 / 0.8198	0.8457 / 0.7254	0.9891 / 0.9947	0.8634 / 0.7826
0.40	0.8821 / 0.8484	0.8555 / 0.8184	0.8408 / 0.7184	0.9897 / 0.9952	0.8608 / 0.7781
0.50	0.8807 / 0.8471	0.8540 / 0.8170	0.8355 / 0.7114	0.9901 / 0.9956	0.8578 / 0.7735
0.60	0.8796 / 0.8455	0.8526 / 0.8153	0.8298 / 0.7044	0.9906 / 0.9959	0.8546 / 0.7689
0.70	0.8780 / 0.8441	0.8509 / 0.8135	0.8234 / 0.6971	0.9911 / 0.9961	0.8510 / 0.7640

Table 7: Impact of decision threshold adjustments on model performance across In-Generation and Cross-Generation datasets. Lowering the inference threshold significantly improves Sensitivity and F1 scores while maintaining a highly robust Specificity (> 0.98). This demonstrates the model’s flexibility to adapt its operating point based on application-specific tolerance for false positives.

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1019 rameters, how they were chosen, type of optimizer) necessary to understand the results?

1020 Answer: [Yes]

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1022 sults are thoroughly documented. The core experimental setting, including the dataset splits,
1023 choice of hyperparameters, evaluation metrics, and hardware specifications is described in
1024 the main text (Section 6.2). Additional exhaustive details, including model training protocols
1025 (e.g., LoRA configurations, optimizer settings) are provided in the Appendices and are fully
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1032 material.

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1034 Question: Does the paper report error bars suitably and correctly defined or other appropriate
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